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Can pyrolysis and composting of sewage sludge reduce the release of traditional and emerging pollutants in agricultural soils? Insights from field and laboratory investigations

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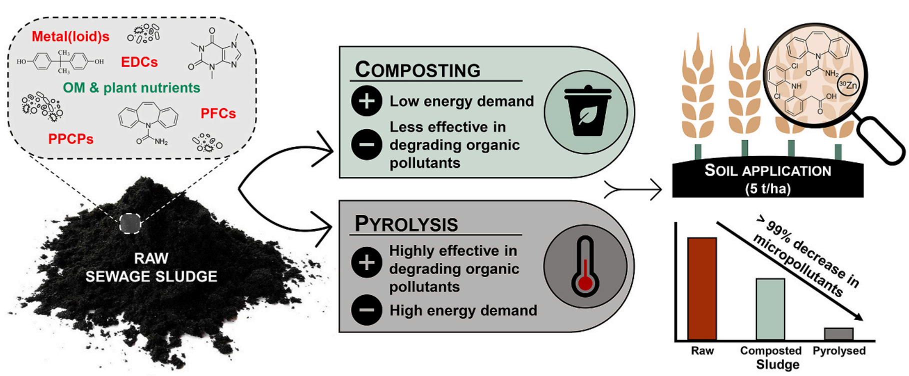
HIGHLIGHTS

- Pyrolysis removed endocrine-disrupting chemicals and PFAS from sludge.
- Leaching of metal(loid)s from sludges was significantly influenced by the treatments.
- Diclofenac and carbamazepine were detected only in wheat seeds after application.
- Telmisartan and triclosan were the most abundant emerging pollutants in sludges.

GRAPHICAL ABSTRACT



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ABSTRACT

The potential extractability, crop uptake, and ecotoxicity of conventional and emerging organic and metal(loid) contaminants after the application of pre-treated (composted and pyrolysed) sewage sludges to two agricultural soils were evaluated at field and laboratory scale. Metal(loid) extractability varied with sludge types and pre-treatments, though As, Cu, and Ni decreased universally. In the field, the equivalent of 5 tons per hectare of both composted and pyrolysed sludges brought winter wheat grain metal(loid) concentrations below statutory limits. Carbamazepine, diclofenac, and telmisartan were the only detected organic pollutants in crops decreasing in order of root > shoot > grains, whilst endocrine-disrupting chemicals, such as bisphenol A and

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